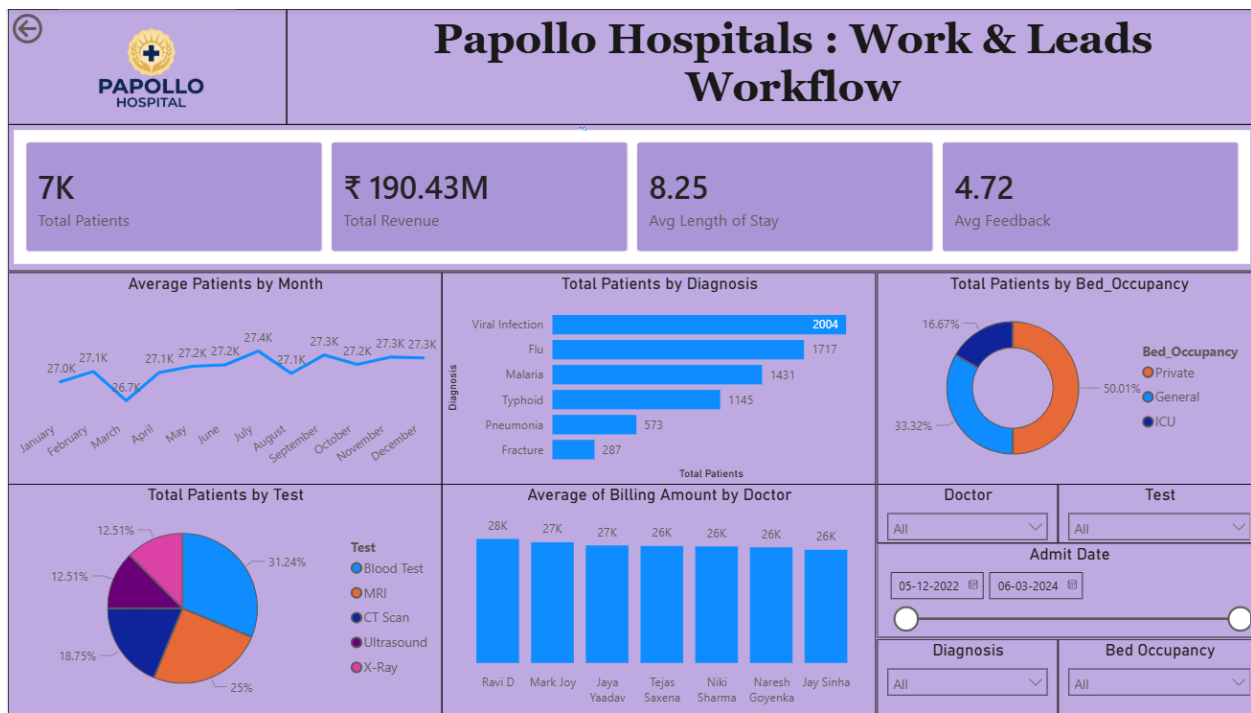


Papollo Hospitals – Work & Leads Workflow Analysis Report

Prepared Using Power BI Dashboard Analytics

Prepared By: Viraj Chavan
Tool Used: Microsoft Power BI
Report Type: Hospital Operations & Performance Dashboard Analysis
Objective: To analyze patient flow, revenue generation, diagnosis trends, bed occupancy utilization, testing distribution, and doctor-wise billing performance using interactive business intelligence dashboards.



Introduction

Healthcare organizations generate massive amounts of operational and patient-related data daily. Proper analysis of this data helps hospitals improve patient care, optimize resource utilization, increase operational efficiency, and make strategic business decisions.

This Power BI dashboard titled “**Papallo Hospitals: Work & Leads Workflow**” was developed to provide a complete analytical overview of hospital operations. The dashboard integrates multiple hospital performance indicators into a single visual platform for easier monitoring and decision-making.

The report focuses on the following major operational areas:

- Total patient management
- Revenue analysis
- Patient diagnosis trends
- Bed occupancy distribution
- Average patient stay duration
- Test utilization patterns
- Doctor-wise billing analysis
- Monthly patient trend analysis

The dashboard uses interactive visualizations such as:

- KPI Cards
- Line Charts
- Bar Charts
- Pie Charts
- Donut Charts
- Slicers & Filters

This analysis helps hospital management identify operational strengths, bottlenecks, patient trends, and areas requiring improvement.

Dashboard Overview

The dashboard provides a summarized operational snapshot of Papollo Hospitals. The key performance indicators (KPIs) displayed at the top of the dashboard include:

KPI	Value	Meaning
Total Patients	7K	Total patients handled
Total Revenue	₹190.43M	Total revenue generated
Average Length of Stay	8.25 Days	Average hospitalization duration
Average Feedback	4.72	Overall patient satisfaction score

Key Observations

1. The hospital handled approximately **7,000 patients**, indicating strong operational capacity and patient inflow.
2. The hospital generated around **₹190.43 Million** in revenue, reflecting healthy financial performance.
3. The average patient stay of **8.25 days** indicates moderate hospitalization duration and suggests that many cases may involve serious or medium-complexity treatments.
4. The average patient feedback score of **4.72/5** reflects high patient satisfaction and quality healthcare service delivery.

The dashboard is highly interactive and allows filtering by:

- Doctor
- Test Type
- Diagnosis
- Bed Occupancy
- Admission Date

These filters help management perform deeper operational analysis and monitor specific departments or time periods.

Monthly Patient Trend Analysis

The line chart in the dashboard represents the monthly patient trends throughout the year.

Monthly Data Insights

Month	Average Patients
January	27.0K
February	27.1K
March	26.7K
April	27.1K
May	27.2K
June	27.2K
July	27.4K
August	27.1K
September	27.3K
October	27.2K
November	27.3K
December	27.3K

Analysis

- Patient inflow remains relatively stable throughout the year.
- March shows a slight decline, possibly due to seasonal factors or reduced admissions.
- July recorded the highest patient volume, indicating peak operational activity.
- The hospital maintains consistent patient engagement without major fluctuations.

Business Implications

- Stable patient inflow helps in better workforce and inventory planning.
- Seasonal variations can be analyzed further for staffing optimization.
- High patient consistency indicates strong brand trust and service demand.

Diagnosis-Wise Patient Analysis

The horizontal bar chart highlights the number of patients treated under different diagnoses.

Diagnosis	Total Patients
Viral Infection	2004
Flu	1717
Malaria	1431
Typhoid	1145
Pneumonia	573
Fracture	287

Analysis

1. **Viral Infection** is the most common diagnosis category, accounting for the highest number of patients.
2. **Flu and Malaria** also contribute significantly to hospital admissions, indicating seasonal or environmental health challenges.
3. Lower fracture cases indicate that Papollo Hospitals may focus more on general medicine and infection treatment rather than trauma care.

Insights

- The hospital should ensure adequate staffing and medicine inventory for viral and flu-related treatments.
- Preventive healthcare campaigns can reduce seasonal infection burden.
- Data-driven diagnosis monitoring helps improve emergency preparedness.

Recommendations

- Strengthen infection control systems.
- Increase awareness programs during seasonal outbreaks.
- Maintain sufficient ICU and isolation bed availability during peak infection periods.

Bed Occupancy Analysis

Total Patients by Bed Occupancy

The donut chart represents patient distribution across different bed categories.

Bed Type	Percentage
----------	------------

Private	50.01%
---------	--------

General	33.32%
---------	--------

ICU	16.67%
-----	--------

Analysis

- More than half of the patients prefer **Private Beds**, indicating higher demand for premium healthcare services.
- General beds account for approximately one-third of occupancy.
- ICU occupancy is lower but critical due to specialized care requirements.

Operational Insights

- High private bed utilization contributes positively to hospital revenue.
- ICU occupancy should always be monitored carefully because of limited availability and high treatment dependency.
- The hospital demonstrates balanced utilization across healthcare service categories.

Recommendations

- Expand private room capacity to maximize profitability.
- Optimize ICU resource allocation for emergency preparedness.
- Introduce dynamic bed management systems for improved efficiency

Test Utilization Analysis

The pie chart displays the percentage distribution of medical tests conducted.

Test Type	Percentage
Blood Test	31.24%
MRI	25%
CT Scan	18.75%
Ultrasound	12.51%
X-Ray	12.51%

Analysis

- Blood tests are the most frequently conducted diagnostic procedure.
- MRI and CT scans also contribute significantly, indicating advanced diagnostic service usage.
- Ultrasound and X-Ray usage remain moderate.

Insights

- Diagnostic services play a major role in hospital revenue generation.
- High imaging demand indicates the importance of radiology infrastructure.
- Blood testing demand reflects strong routine and infection-based diagnostics.

Recommendations

- Upgrade diagnostic lab efficiency to reduce turnaround time.
- Expand imaging equipment capacity during peak demand periods.
- Implement AI-assisted diagnostics for operational efficiency.

Doctor-Wise Billing Analysis

The bar chart analyzes average billing generated by different doctors.

Doctor	Average Billing
Ravi D	28K
Mark Joy	27K
Jaya Yaadav	27K
Tejas Saxena	26K
Niki Sharma	26K
Naresh Goyenka	26K
Jay Sinha	26K

Analysis

- Dr. Ravi D generated the highest average billing amount.
- Billing values among doctors are relatively balanced, indicating fair workload distribution.
- Minor variations suggest consistent operational performance among medical professionals.

Insights

- Balanced billing indicates effective doctor allocation strategies.
- High-performing doctors can mentor others for operational optimization.
- Billing analytics can help in incentive planning and resource allocation.

Recommendations

- Introduce doctor performance dashboards.
- Analyze patient satisfaction alongside billing performance.
- Use analytics for appointment scheduling optimization.

Strategic Business Insights & Recommendations

Strategic Insights

The dashboard provides several important business and operational insights for Papollo Hospitals:

1. Strong Revenue Performance

The hospital generated ₹190.43M in revenue, demonstrating financial stability and effective operational management.

2. High Patient Satisfaction

An average feedback score of 4.72 reflects excellent patient care quality and hospital service standards.

3. Infection-Based Patient Dominance

Most admissions are related to viral infections and flu, highlighting the need for seasonal preparedness.

4. Premium Healthcare Demand

Private bed occupancy exceeding 50% indicates rising demand for premium healthcare facilities.

5. Stable Patient Flow

Monthly patient trends show consistent inflow, supporting predictable resource planning.

Recommendations

Operational Recommendations

- Improve emergency and infection response systems.
- Expand diagnostic infrastructure.
- Enhance digital patient monitoring systems.

Financial Recommendations

- Increase premium room services for higher profitability.
- Optimize doctor scheduling for efficient billing management.

Technology Recommendations

- Integrate predictive analytics for patient forecasting.
- Implement AI-based operational monitoring systems.

Patient Care Recommendations

- Strengthen preventive healthcare awareness programs.
- Improve discharge management to reduce average stay duration.

Conclusion

The **Papollo Hospitals Work & Leads Workflow Dashboard** successfully demonstrates how business intelligence and data visualization tools can improve healthcare management and decision-making.

Using Power BI, the dashboard transforms complex hospital data into meaningful visual insights that help management monitor operational performance effectively. The analysis highlights strong revenue generation, stable patient inflow, high patient satisfaction, and balanced resource utilization.

The dashboard also identifies critical operational areas such as infection management, bed occupancy optimization, diagnostic service demand, and doctor-wise billing efficiency. These insights support better planning, cost control, resource allocation, and patient care improvement.